

Measuring What Matters: Travel Barriers, Transportation Insecurity, and Perceived Accessibility as Equity-Informed Performance Indicators

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Why We Need Better Measures of Everyday Travel

1 in 4

Americans are unable to regularly and reliably access transportation to meet their daily needs

(FTA, 2023)

19%

of Americans experienced transportation insecurity in the past month — exceeding food insecurity, housing insecurity, and unmet medical needs

(Murphy & Pilkauskas, 2024)

Yet we still rely on system-level metrics (speed, congestion, VMT) that cannot capture these experiences.

How Do We Measure Transportation Performance?

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What Agencies Currently Measure

Safety

Fatalities, serious injuries, crash rate per VMT

Infrastructure Condition

Pavement condition (IRI, cracking, rutting), bridge condition

System Performance

Travel time reliability, freight reliability, peak hour excessive delay, CMAQ emissions

Additional Long Range Transportation Planning indices

Mode split, VMT, level of service, congestion index, transit ridership, revenue miles

System-focused:
How is the infrastructure performing?

What Equity-Informed Measures Could Add

Travel Needs & Barriers

Trips not taken, activities postponed, travel problems encountered (cost, time, safety, comfort)

Transportation Insecurity

Inability to regularly and reliably reach daily needs; skipped destinations, rescheduled appointments

Perceptions of Accessibility

Self-reported ease of reaching activities, satisfaction with access to opportunities

People-focused:
Can travelers meet their daily needs?



Study Objective

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Develop and validate a survey instrument that captures perceived travel experiences across the U.S. population.



Measure

Capture self-reported travel barriers, transport insecurity, and perceived accessibility through a scalable online survey



Compare

Examine how experiential measures differ across population groups (income, disability, car access, geography)



Validate

Test whether survey-based measures predict unmet needs, activity participation, and life satisfaction



Inform Policy

Demonstrate how these indicators can complement existing agency performance frameworks (LRTPs, equity screening)

Data Collection Overview



Online Survey

Administered via Qualtrics panel on U.S. general population



Quota Sampling

Stratified by gender, age, race/ethnicity, and U.S. Census region



Summer 2025

Fielded July 15 – August 15, 2025



54 Questions

Covering 7 key themes; average completion time ~12 minutes



Quality Control

2 attention checks, completion time screening, straight-lining detection



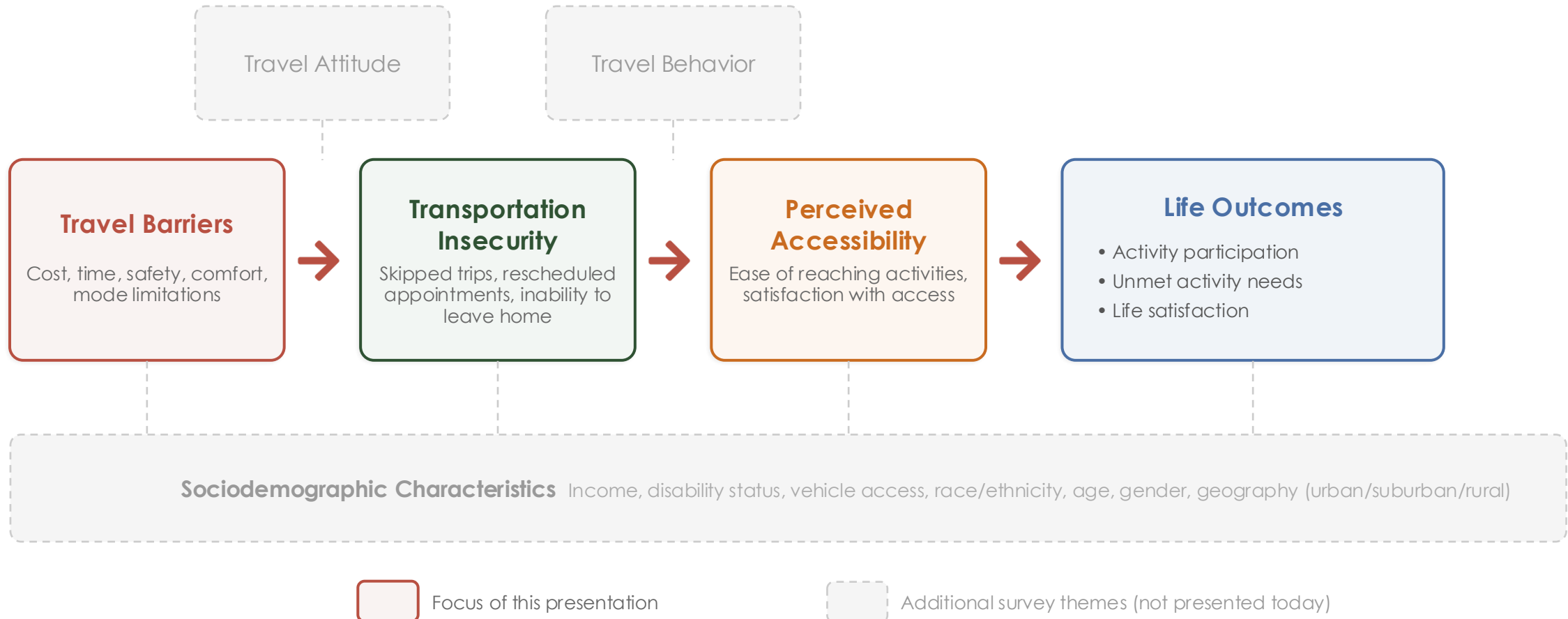
616 Final Sample

670 responses collected; 616 retained after quality screening

Conceptual Framework

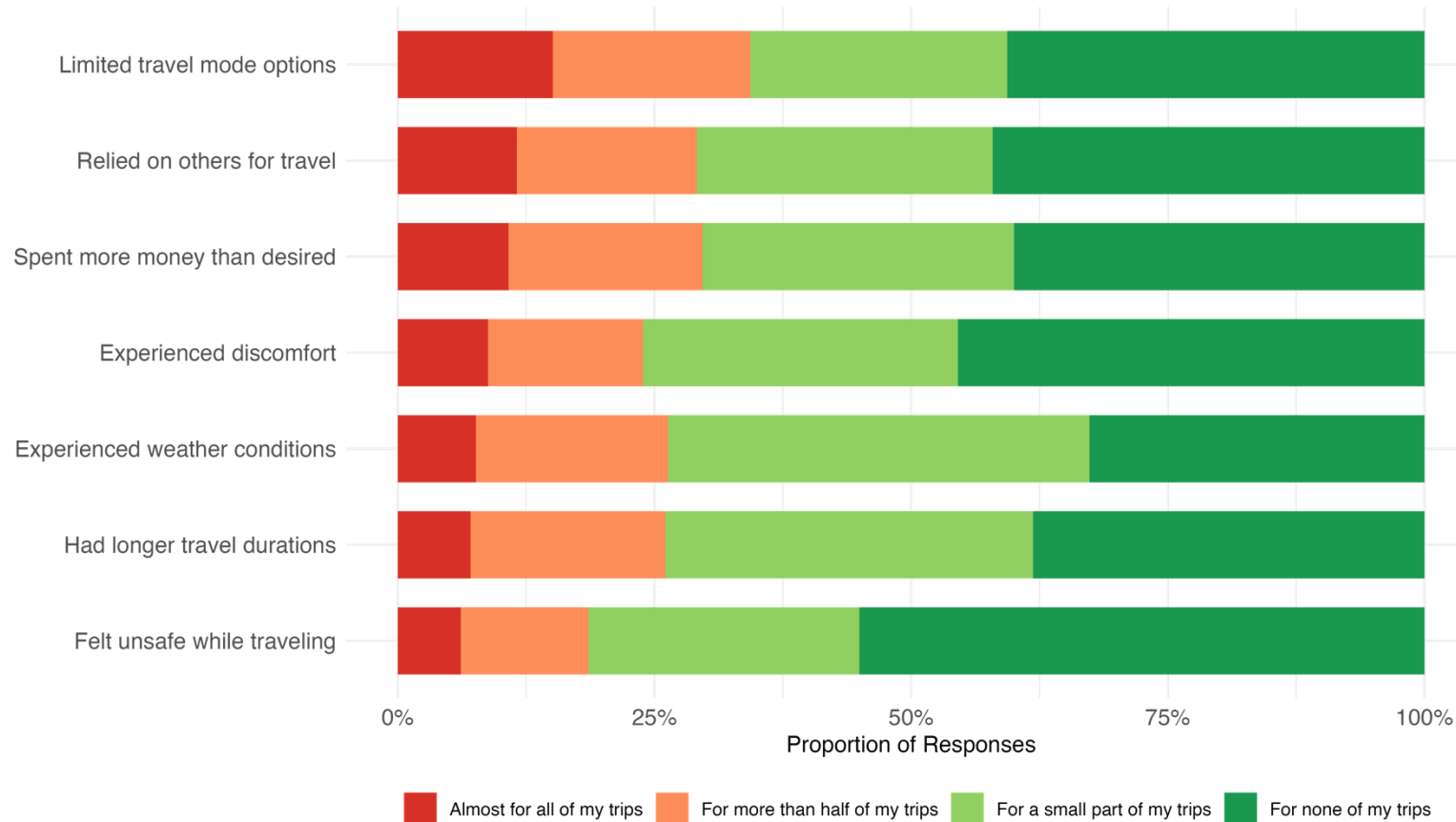
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Constructs highlighted in color are the focus of this presentation; the full survey instrument covers additional themes shown in gray.



Travel Barrier Response

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Travel Barriers Index

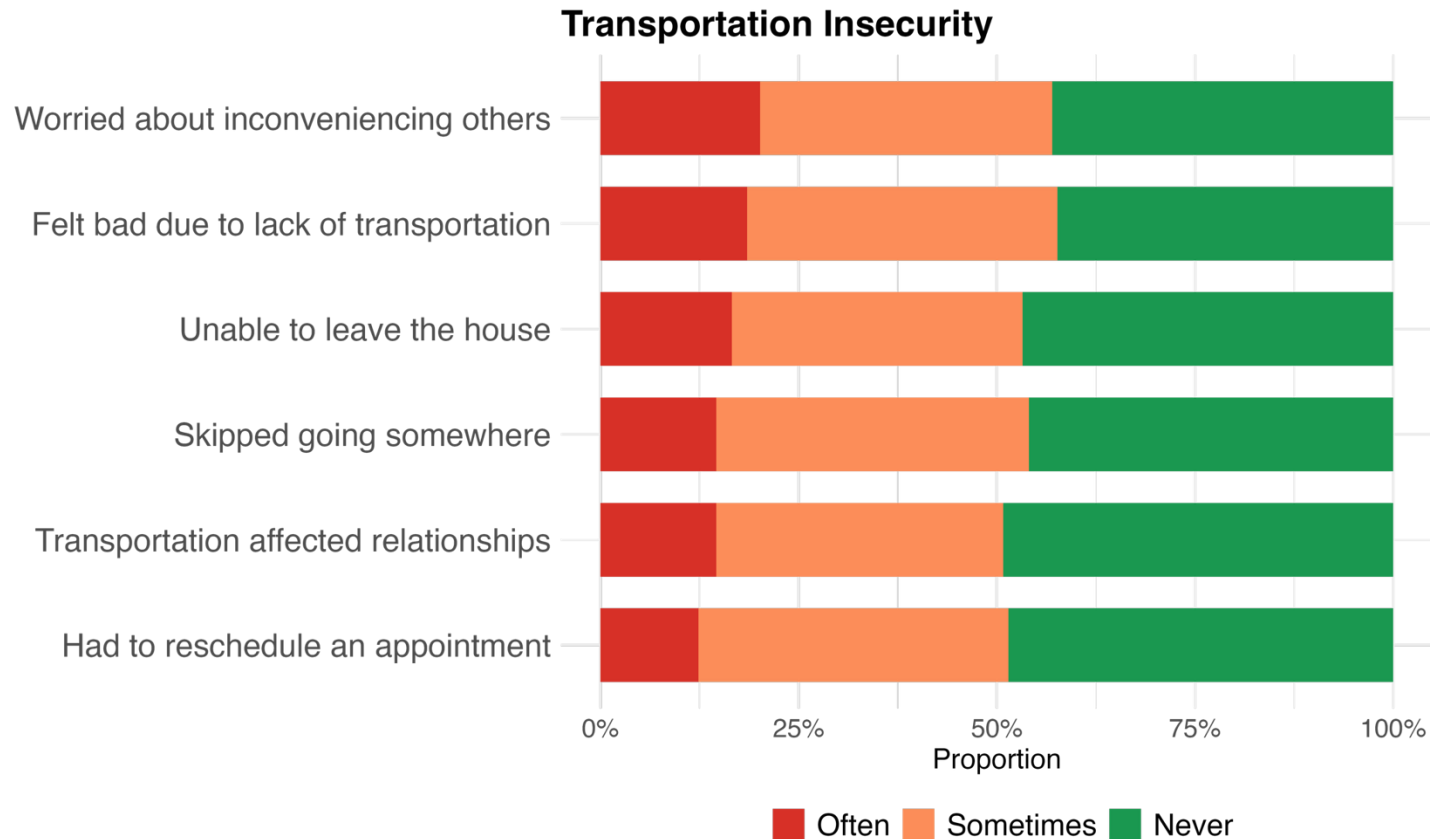
0 = None of my trips
1 = Small part of my trips
2 = More than half of my trips
3 = Almost all of my trips

Sum across 7 items (range 0–21)

0–3: No/Marginal barriers
4–9: Low/Moderate barriers
≥10: High barriers

Transport Insecurity Response

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Transport Insecurity Index

0 = Never, 1 = Sometimes, 2 = Often

Sum across 6 items (range 0–12)

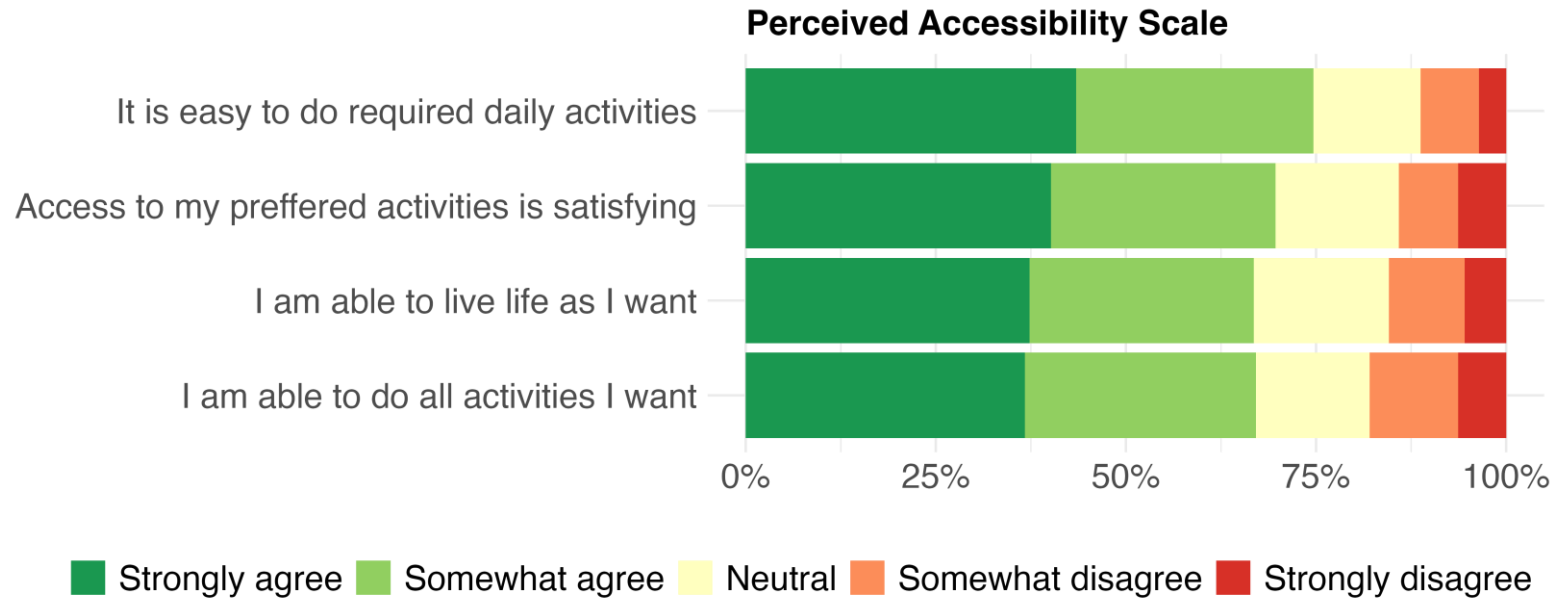
0–1: Secure

2–6: Low/Marginal

7–12: Moderate/High Insecurity

Perceived Accessibility Response

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Perceived Accessibility Index

4-item scale (Lättman et al. 2016)

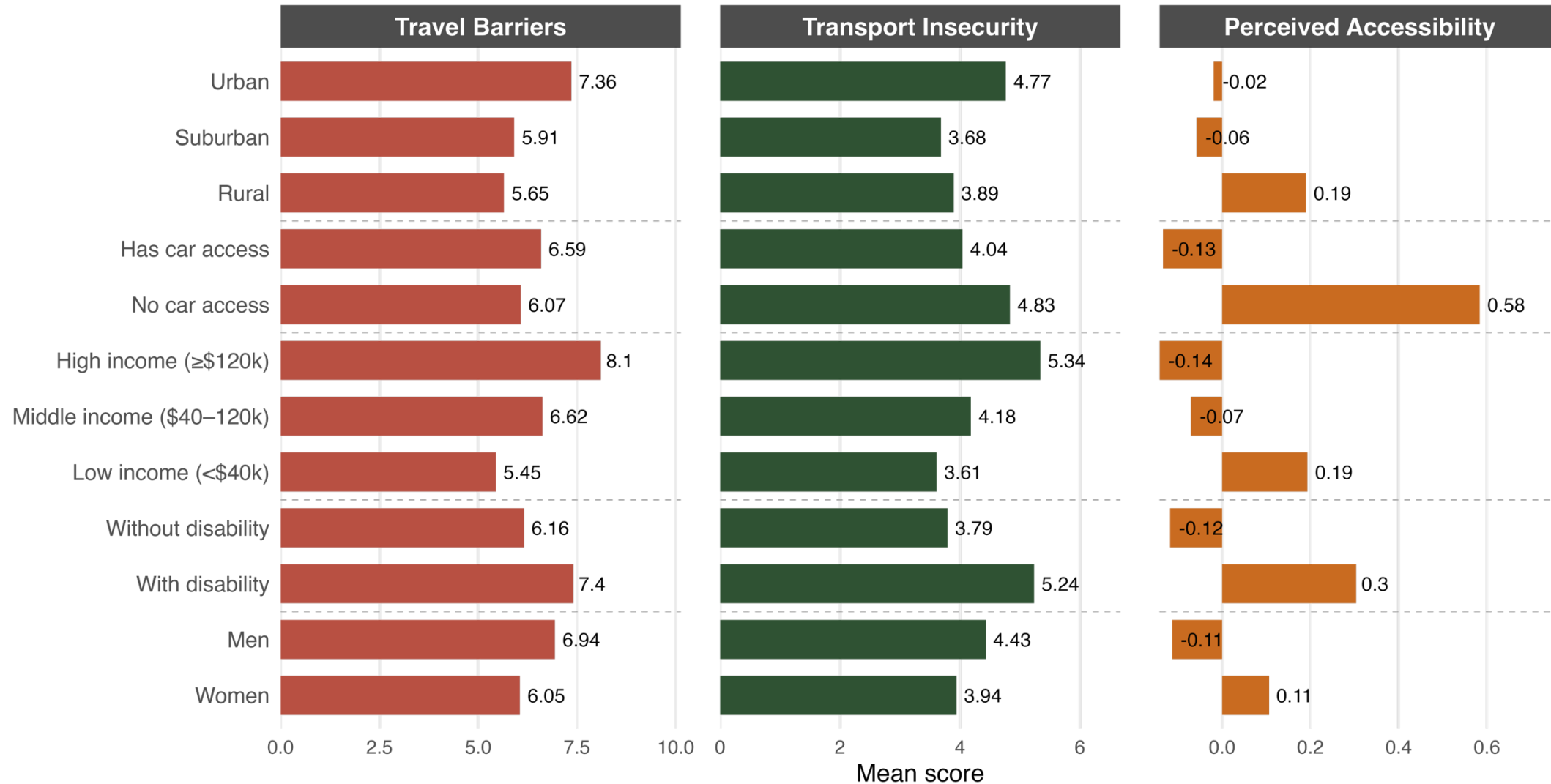
Standardized composite score

Negative = lower perceived access

Zero = sample average

Experiential Measures Across Population Groups

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Validating Experiential Measures

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Do these measures predict real outcomes? ANOVA results by transport insecurity level.

Outcome Measure	Secure (N=214)	Low/Marginal (N=246)	Moderate/High (N=156)	Sig.
# Unique activities in past 7 days	3.52	3.53	3.42	0.924
Unmet Activity Need (% Yes)	16.1%	68.0%	81.9%	<0.001
Life satisfaction (1–10)	6.71	5.89	5.85	<0.001

16.1% → 81.9%: Unmet activity needs increase fivefold as transport insecurity rises. Life satisfaction also decline significantly.

Policy Implications

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How can transport agencies integrate experiential measures into practice?



Embed in Existing Surveys

Transport insecurity and perceived accessibility questions can be added to household travel surveys and community engagement instruments that MPOs already conduct.



Strengthen Equity Screening

Use experiential measures to identify communities where modeled accessibility looks adequate but residents report high insecurity or low perceived access.



Evaluate Project Outcomes

Deploy before-and-after surveys to assess whether investments actually reduce missed trips, unmet needs, and travel disruptions for affected populations.



Complement, Not Replace

These measures do not replace existing performance metrics — they add a traveler-centric layer that validates whether system improvements translate to better outcomes.

A scalable, low-cost approach to performance-based planning that aligns accessibility goals with lived mobility outcomes.

Limitations & Next Steps

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Limitations

- 1 Sample size for subgroups**
Only 26% experienced moderate-to-high transport insecurity, limiting subgroup analyses
- 2 Cross-sectional design**
Single survey wave cannot establish causality between constructs
- 3 Online panel bias**
Qualtrics panels may underrepresent underserved populations (e.g., limited internet access)
- 4 Subjectivity of self-report**
Perceived measures may reflect individual expectations as well as actual conditions

Next Steps

- 1 Round 2 national survey**
Larger representative sample with refined question framing based on Round 1 findings
- 2 Validity & reliability testing**
Confirmatory factor analysis and test-retest reliability at national and local scales
- 3 Agency integration pilot**
Partner with MPOs/DOTs to test how these metrics can be incorporated into LRTPs and project prioritization
- 4 Construct-specific validation**
Examine how travel barriers and perceived accessibility categories independently predict life outcomes

Thank You!



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Extra Slides for Q/A

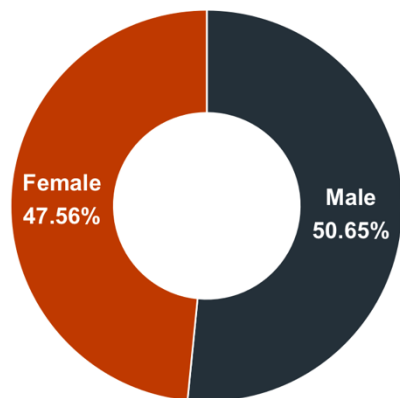
Measuring Transport Insecurity

Thinking about the past 30 days:

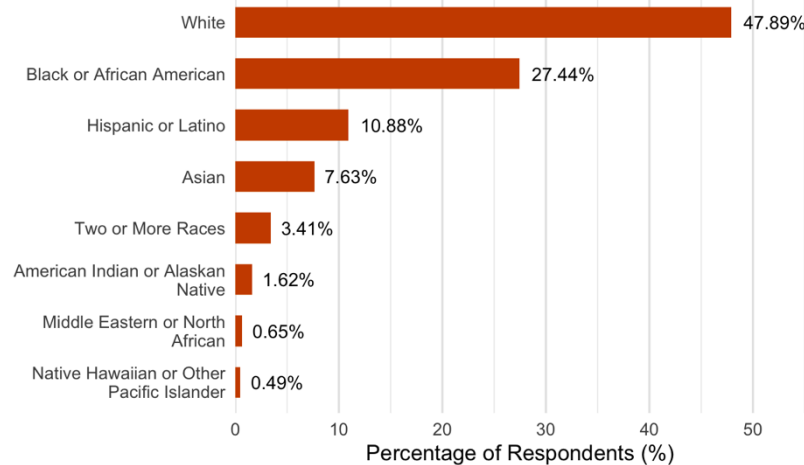
	Never	Sometimes	Often
How often did you have to reschedule an appointment because of a problem with transportation?	☐	☐	☐
How often did you skip going somewhere because of a problem with transportation?	☐	☐	☐
How often were you unable to leave the house when you wanted to because of a problem with transportation?	☐	☐	☐
How often did you feel bad because you did not have the transportation you needed?	☐	☐	☐
How often did you worry about inconveniencing your friends, family, or neighbors because you needed help with transportation?	☐	☐	☐
How often did problems with transportation affect your relationships with others?	☐	☐	☐

Sample Distribution

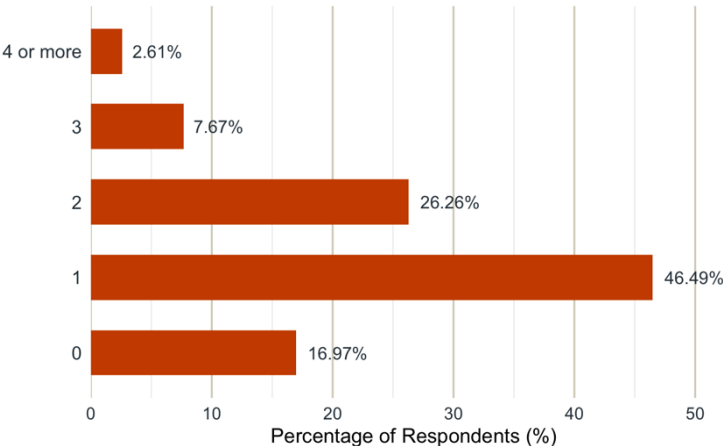
Gender Distribution of Respondents



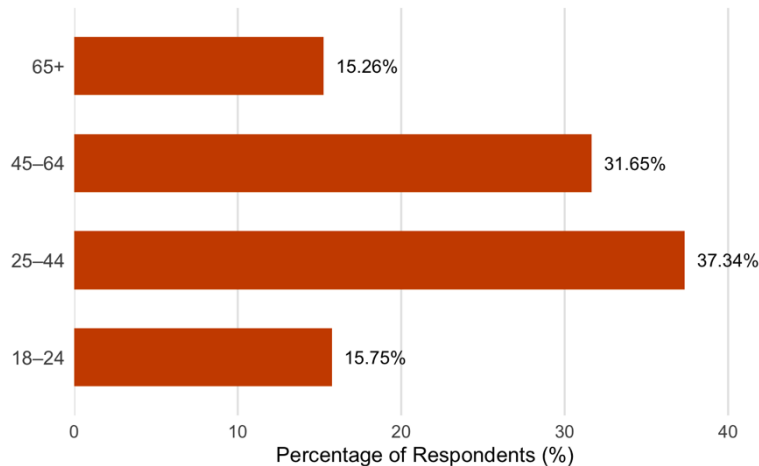
Race/Ethnicity Distribution of Respondents



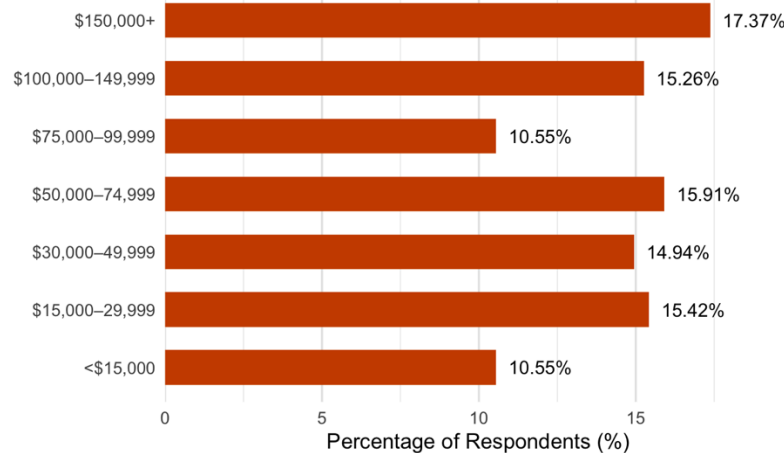
Household Vehicle Ownership



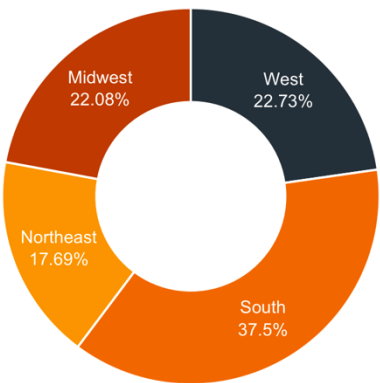
Age Distribution of Respondents



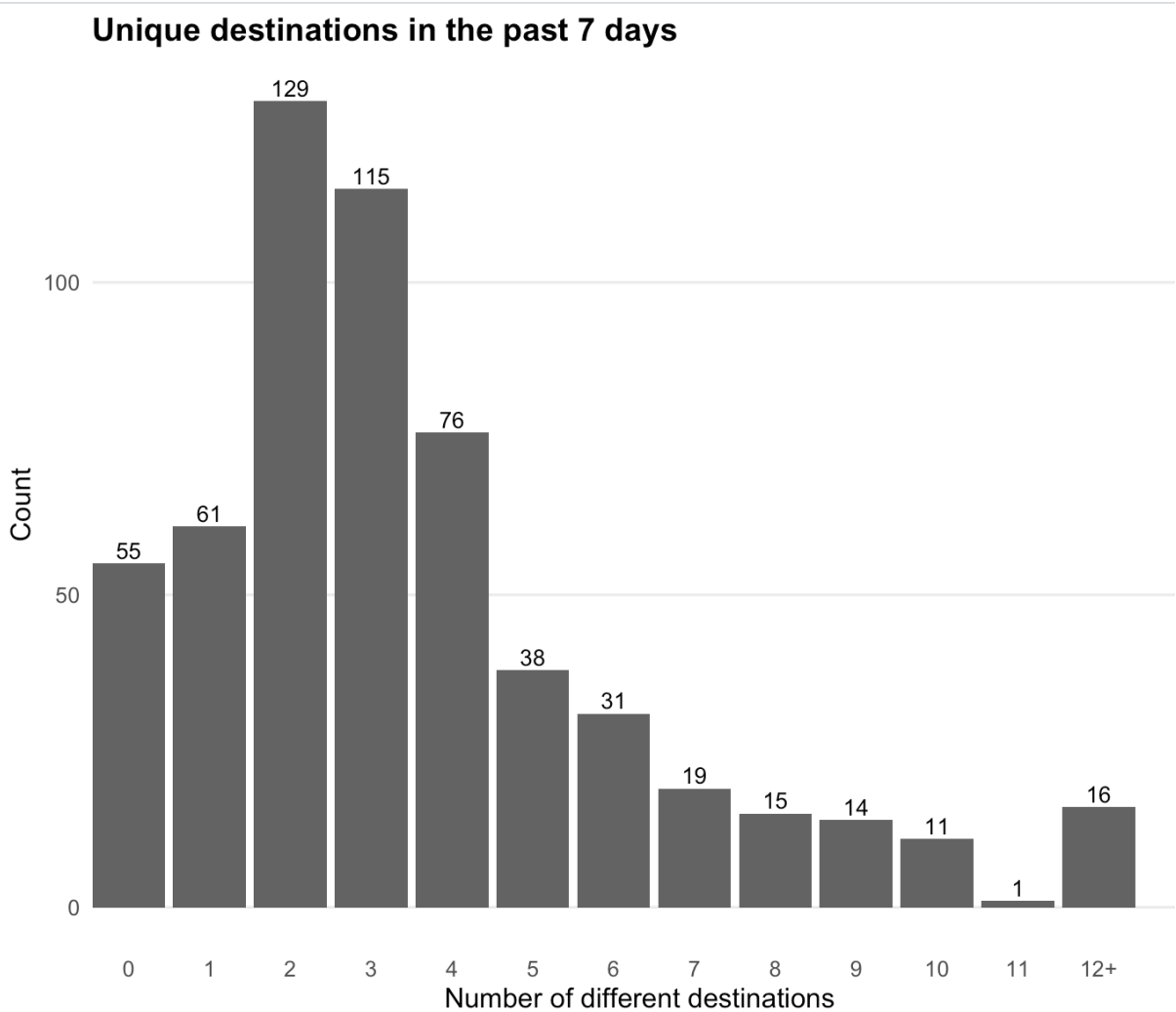
Household Income Distribution



Region of Residence



Descriptive Statistics



Descriptive Statistics

